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بازت اول :

a)

$$\begin{bmatrix} 4\omega_1 - 6 \\ 2\omega_2 - 4 \end{bmatrix}$$

b)

$$\begin{bmatrix} 4\omega_1^* - 6 \\ 2\omega_2^* - 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\omega_1^* = 1 \quad \omega_2^* = 2$$

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

c)

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \alpha \begin{bmatrix} -6 \\ -4 \end{bmatrix}$$

$$\alpha = 0.15 \rightarrow \begin{bmatrix} 0.15 \\ 0.15 \end{bmatrix} \quad \alpha = 0.15 \rightarrow \begin{bmatrix} 1.2 \\ 1.8 \end{bmatrix}$$

d)

$$\text{نقطه مبدأ} \rightarrow \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\vec{w}_A = \begin{bmatrix} 0.15 \\ 0.15 \end{bmatrix}$$

$$\vec{w}_B = \begin{bmatrix} 1.2 \\ 1.8 \end{bmatrix}$$

فاصله از A : 1.93

فاصله از B : 1.22

۱. A : $\alpha = 0.15$ قدم های کوچکی بردارد و B : $\alpha = 0.12$ قدم های بزرگتری دارد

قدم B هنوز بزرگتر است که از نقطه مبدأ شروع به حرکت کند بنابراین در هر قدم به سمت نقطه مبدأ حرکت می کند و توان ندارد.

$$a) \hat{y} = \vec{w}^T \vec{x} + b$$

$$\hat{y} = w_1 x_{i1} + w_2 x_{i2} + b$$

$$\vec{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \quad \vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$J: MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$\vec{w}^{(t+1)} = \vec{w}^{(t)} - \alpha \nabla_{\vec{w}} J$$

$$b^{(t+1)} = b^{(t)} - \alpha \frac{dJ}{db}$$

$$\nabla_{\vec{w}} J = -\frac{1}{n} X^T (\vec{y} - \hat{\vec{y}})$$

$$\frac{dJ}{db} = -\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)$$

$$b) w_1 = 0 \quad w_2 = 0 \quad b = 0 \quad (1^{st} \text{ iteration})$$

$$x_{121} \quad x_{122} \quad y_{12}$$

$$\hat{y}_{12} = y - \hat{y}_{12}$$

$$x_{121} \quad x_{122} \quad y_{12} \quad (2^{nd} \text{ iteration})$$

$$\hat{y}_{12} = y - \hat{y}_{12}$$

$$x_{121} \quad x_{122} \quad y_{12} \quad (3^{rd} \text{ iteration})$$

$$\hat{y}_{12} = y - \hat{y}_{12}$$

: Gradient descent الجري ١٦

$$\frac{\partial J}{\partial w_1} = -\frac{1}{n} \sum e_i x_{i1} = -1$$

$$\frac{\partial J}{\partial w_2} = -0.3333$$

$$\frac{\partial J}{\partial b} = -0.9999$$

$$w_1^{(1)} = 0 - 0.1(-1) = 0.1$$

$$w_2^{(1)} = 0 - 0.1(-0.3333) = 0.0333$$

$$b^{(1)} = 0.9999$$

$$w_1 = 0.1, w_2 = 0.0333, b = 0.9999 \text{ : Epoch 1}$$

Epoch 2 :

$$w_1 = 0.1, w_2 = 0.0333, b = 0.9999$$

$$\text{Line 1 : } x_1 = 1, x_2 = 0, \hat{y} = 1.0333$$

$$e = 0.0333$$

$$\text{Line 2 : } x_1 = 2, x_2 = 1, \hat{y} = 2.1 e = 2.1$$

$$\text{Line 3 : } x_1 = 0, x_2 = 1, \hat{y} = 1.1 e = 1.1$$

$$\frac{\partial J}{\partial w_1} = -3.1222 / \frac{\partial J}{\partial w_2} = -2.1444 / \frac{\partial J}{\partial b} = -0.0222$$

$$w_1^{(2)} = 1.1222, w_2^{(2)} = 0.1, b^{(2)} = 0.9778$$

$$\hat{y} = 1.1222 x_1 + 0.1 x_2 + 0.9778$$

$$MSE_{\text{Training}} = 1.1444, MSE_{\text{Test}} = 0.1222$$